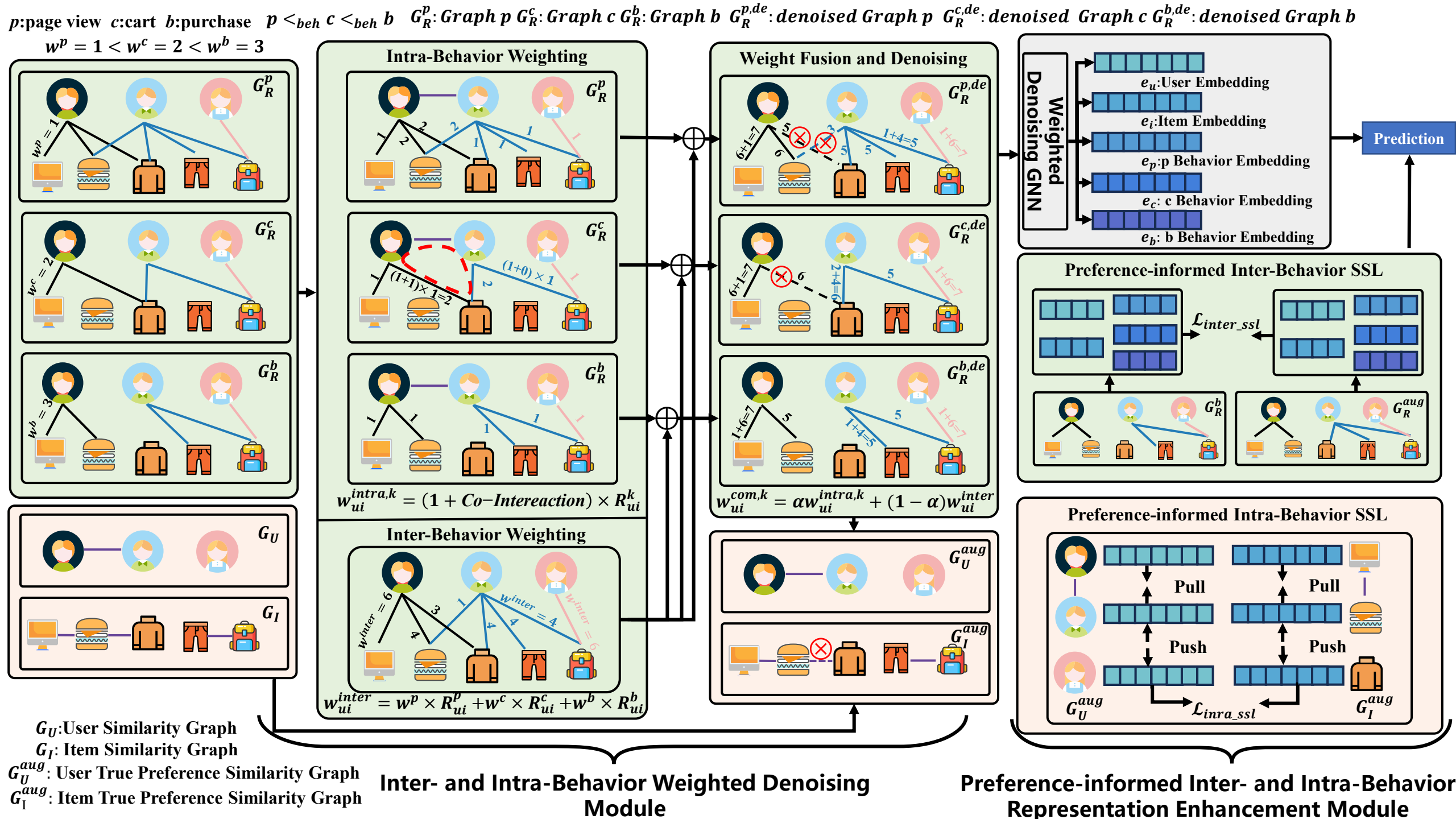
p:page view c:cart b:purchase p <beh c <beh b
G_R^p: Graph p G_R^c: Graph c G_R^b: Graph b $G_R^{p,de}$: denoised Graph p $G_R^{c,de}$: denoised Graph c $G_R^{b,de}$: denoised Graph b
$w^p = 1 < w^c = 2 < w^b = 3$
Intra-Behavior Weighting
Weight Fusion and Denoising
Weighted Denoising GNN
e_u:User Embedding
e_i:Item Embedding
e_p:p Behavior Embedding
e_c: c Behavior Embedding
e_b: b Behavior Embedding
Prediction
Preference-informed Inter-Behavior SSL
$\mathcal{L}_{inter_ssl}$
$w_{ui}^{intra,k} = (1 + Co\text{-}Intereaction) \times R_{ui}^k$
$w_{ui}^{com,k} = \alpha w_{ui}^{intra,k} + (1-\alpha) w_{ui}^{inter}$
Inter-Behavior Weighting
$w_{ui}^{inter} = w^p \times R_{ui}^p + w^c \times R_{ui}^c + w^b \times R_{ui}^b$
Preference-informed Intra-Behavior SSL
Pull
Push
$\mathcal{L}_{inra_ssl}$
G_U:User Similarity Graph
G_I: Item Similarity Graph
G_U^{aug}: User True Preference Similarity Graph
G_I^{aug}: Item True Preference Similarity Graph
Inter- and Intra-Behavior Weighted Denoising Module
Preference-informed Inter- and Intra-Behavior Representation Enhancement Module